

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1699

Roanoke Rapids 138 kV
November 29, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1699. The Interconnection Customer proposes a 101.0 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on March 24, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 459 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,716,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MECKLENBURG POWER STATION-BEAR 345kV	CLOVER-BEAR 345kV	113.8	2,827	17.6
WING LANE-SECO 138kV Line	MITCHELL DAM-SKAGS ISLAND JCT 2 345kV	111.3	413	9.9
SECO-HARVARD 138kV	MITCHELL DAM-SKAGGS ISLAND JUNCTION 2 345kV Line	109.5	524	9.9
WING LANE-SPRING STREET SUBSTATION 138kV	MITCHELL DAM-SKAGGS ISLAND JCT 2 345kV	106.5	309	7.5
FRUITRIDGE & S. LAND PARK-REDBED PLAINS WIND FARM 138kV line	MITCHELL DAM-SKAGGS ISLAND JCT 2 345kV	106.4	341	6.2
FRUITRIDGE & S. LAND PARK-CARLIN 138kV	MITCHELL DAM-SKAGGS ISLAND JUNCTION 2 345kV	106.0	385	6.2
BEAR ISLAND-CHANCELLOR ROW 345kV	LEAVITT-BIRCHWOOD 345kV	102.9	2,328	24.1
N TOWER TAP-DPC ELROY 138kV	MCCARTNEY-N TOWR TAP 345kV L	102.9	661	6.2
LEAVITT-BIRCHWOOD 345kV	BEAR ISLAND-DUNN SIDING 345kV line	100.8	3,031	24.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MECKLENBURG POWER STA-BEAR 345kV line	Rebuild line with higher-capacity conductor	\$113,376,000	1% (\$1,045,000)
WING LANE-SECO 138kV Line	Rebuild line with higher-capacity conductor	\$2,235,000	1% (\$30,000)
HARVARD-SECO 138kV	Reconductor line	\$1,223,000	1% (\$17,000)
WING LANE-SPRING STREET SUBSTATION 138kV	Reconductor line	\$765,000	2% (\$13,000)
FRUITRIDGE & S. LAND PARK-REDBED PLAINS WIND FARM 138kV	Reconductor line	\$2,664,000	1% (\$28,000)
CARLIN-FRUITRIDGE & S. LAND PARK 138kV L	Reconductor line	\$1,090,000	1% (\$11,000)
BEAR ISLAND-CHANCELLOR ROW 345kV	Reconductor line	\$14,673,000	1% (\$126,000)
N TOWER TAP-DPC ELROY 138kV	Reconductor line	\$1,948,000	2% (\$31,000)
LEAVITT-BIRCHWOOD 345kV	Reconductor line	\$50,833,000	1% (\$415,000)
	Total allocated to project		\$1,716,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.